PRODUCT TEARDOWN

Park+

Utility-to-Commerce Flow

Ajay Mahadik

Product Management Portfolio

Overview

MISSION

Digitize offline car ownership. Convert fragmented tasks (parking, challans, FASTag) into a unified ecosystem with automated tracking via RC Number.

VALUE PROPOSITION

Zero-Friction Entry

Automated tracking via RC Number — no manual data entry

Offline Convenience

Seamless boom barrier access at societies and malls

Unified Payments

Centralized payments for all vehicle compliance tasks

REVENUE STREAMS

- **Transactional**: Convenience fees on challans, FASTag recharges: **# of transactions * Avg. convenience fee**
- **SaaS/Subscription**: Daily car wash, society gate management: **# of businesses/Societies * # of subscriptions * Avg. fee**
- **Marketplace**: Lead generation for insurance & accessories: **# of businesses * Avg actions/user * Cost per action**

COST STRUCTURE

- **Hardware deployment (RFID gates)**; **# of societies * (Avg deployment cost + Avg maintenance)**
- **API costs** (Vahan, NPCI infrastructure) : **# of API requests * Cost per API request**
- **Marketing & partner acquisition**: **# of marketplace partners * Avg partner acquisition cost (PAC)**

KEY PARTNERS & ACTORS

- Mall Operations / Housing Societies
- Government (RTO/NPCI)
- Insurance Companies
- Customers (owning cars)
- Service Providers / Brands

Business Model

REVENUE STREAMS

- Transactional: Convenience fees on challans, FASTag recharges
- SaaS/Subscription: Daily car wash, society gate management
- Marketplace: Lead generation for insurance & accessories

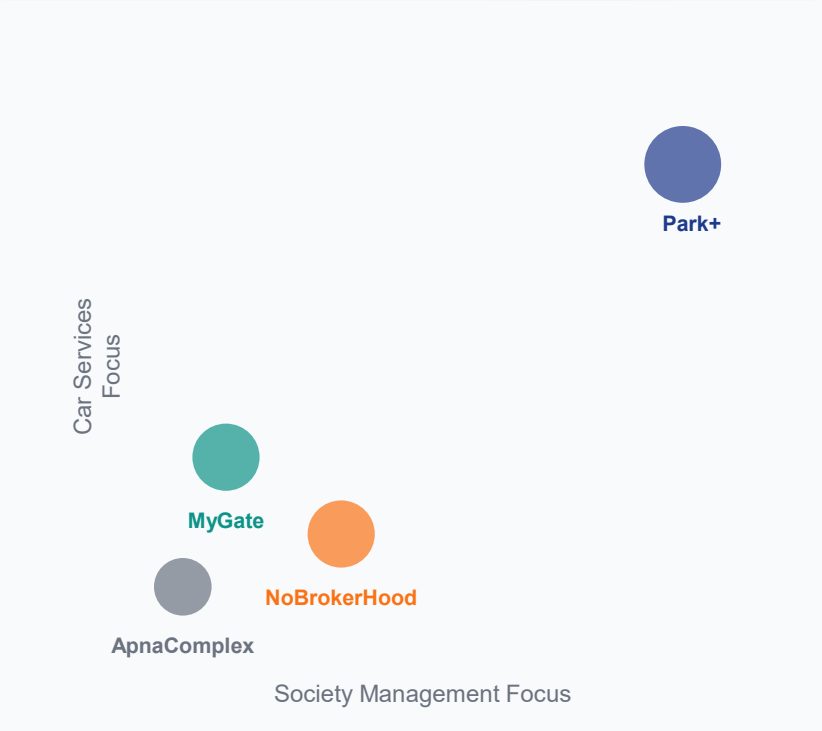
COST STRUCTURE

- Hardware deployment (RFID gates)
- API costs (Vahan, NPCI infrastructure)
- Marketing & partner acquisition

KEY PARTNERS

- RWA / Mall Operations
- Government (RTO/NPCI)
- Insurance Companies

COMPETITIVE LANDSCAPE



User Personas

AMIT VERMA

THE UTILITY SEEKER • 34 • PUNE

LIFESTYLE

IT professional, works long hours. Uses car primarily for weekend trips and daily commute. Prefers quick, no-frills digital solutions.

PERSONALITY TRAITS



GOALS

- Quick FASTag recharges (under 30 sec)
- Instant challan clearance without RTO visit
- Zero manual data entry via RC auto-fetch
- Minimal notifications — only critical alerts

PAIN POINTS

- Hates aggressive cross-sell during high-stress tasks
- Forgets app exists until emergency hits
- Frustrated by irrelevant promotional notifications

PRIYA SINGH

THE ECOSYSTEM ADOPTER • 29 • BANGALORE

LIFESTYLE

Marketing manager, works hybrid. Uses car daily for office and social events. Lives in gated society, values convenience and subscriptions.

PERSONALITY TRAITS



GOALS

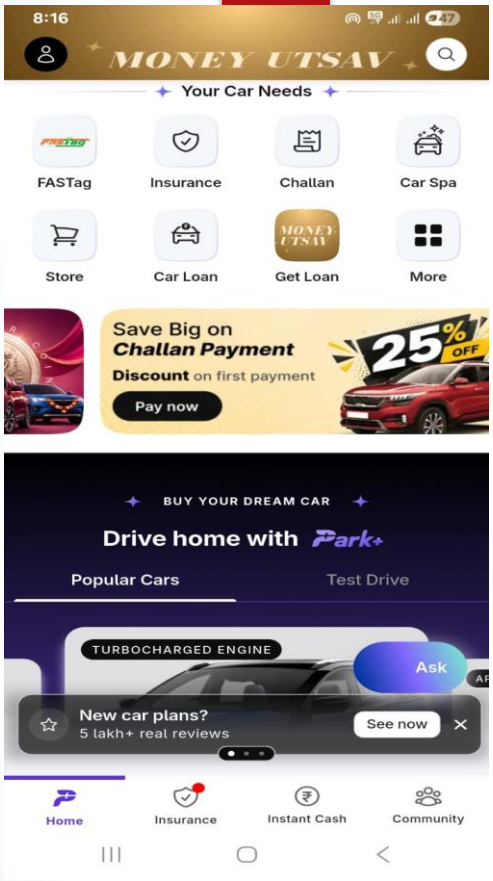
- Automated society gate entry — no manual RFID
- Subscription-based car cleaning
- Proactive reminders for insurance, PUC expiry
- Access to exclusive deals and loyalty rewards

PAIN POINTS

- Haggling with local car cleaners — wants fixed pricing
- Forgets insurance renewal — last-minute panic
- Cluttered dashboard — hard to find core utility

User Journey Mapping

1. Trigger	2. Utility Action	3. The Pivot	4. Reaction	5. Post-Utility
ACTIONS Receives low FASTag SMS at toll booth. Opens app while waiting in queue.	**ACTIONS** Completes 1-click UPI recharge. Payment confirmed in 8-10 seconds.	**ACTIONS** Post-payment UI pushes car wash subscription popup immediately.	**ACTIONS** Dismisses popup rapidly to view receipt. Closes app within 3 seconds.	**ACTIONS** Reopens app 2 days later when relaxed. Browses marketplace, subscribes.
THOUGHTS *"Need to recharge fast before boom barrier."*	**THOUGHTS** *"Wow, that was smooth. Zero manual entry needed."*	**THOUGHTS** *"I'm still at the toll, why show me this now?"*	**THOUGHTS** *"Conditioned to close anything that pops up."*	**THOUGHTS** *"Oh, they have a monthly car wash plan. Convenient."*
ANXIOUS	RELIEVED	FRUSTRATED	ANNOYED	OPEN



KEY INSIGHT: The utility flow (stages 1-2) is exceptionally smooth, but the commerce pivot (stage 3) happens at the wrong moment — when the user is still in a high-anxiety state. Moving the cross-sell to stage 5 (post-utility, relaxed state) could significantly increase conversion.

Key Learnings

UI / UX

Cognitive Overload

Mixing high-anxiety tasks (fines, toll) with commerce creates friction.

SO: Separate utility and discovery modes based on user context.

Conditioned Blindness

Users swipe away pop-ups to reach core utility quickly.

SO: Timing matters more than frequency. Pitch when relaxed.

Frictionless Utility

UPI payment flow is exceptionally smooth — 8-second completion.

SO: This is the core differentiator. Protect and enhance it.

TECH

Primary Key Masterclass

Using Vehicle RC as primary key maps all data instantly. Zero manual entry.

SO: This is the moat. Competitors require 5+ fields; Park+ needs 1.

Real-Time Processing

Instant Vahan and NPCI data fetching enables sub-10-second flows.

SO: Speed is the product. Every delay drops conversion.

Predictive Triggers

Automates insurance reminders based on fetched dates. Proactive > reactive.

SO: Predictive features drive subscription conversions.

DESIGN

Cluttered Dashboard

Home screen functions more like ad-billboard than utility panel.

SO: Hierarchy failure. Users can't find FASTag quickly.

Hierarchy Struggle

Balances core utility (top) with marketplace discovery (scroll).

SO: Context-aware UI needed — utility first when pending tasks.

Consistent Branding

Strong visual identity across offline gates and app. RFID mirrors app.

SO: Offline-online consistency builds trust. Competitive advantage.

Recommendations

1. CONTEXT-AWARE SMART DASHBOARD

Priority: HIGH | Effort: MEDIUM | Impact: HIGH

CURRENT: Aggressive cross-selling interrupts high-stress utility tasks.
RECOMMENDATION: Introduce state-machine UI. If user has pending fine/low FASTag, enter 'Utility Mode' — strip all ads. Once resolved, transition to 'Discovery Mode'.
EXAMPLE: User pays challan → UI shows 'All clear!' → 2 sec later, shows car wash renewal prompt → Relaxed user converts better.

2. GRANULAR NOTIFICATION CONTROLS

Priority: HIGH | Effort: LOW | Impact: MEDIUM

CURRENT: Standardized push alerts for marketplace. Users receive promotional alongside critical alerts.
RECOMMENDATION: Allow users to opt out of promotional alerts while keeping critical utility alerts active. Add frequency controls.
EXAMPLE: User settings: 'FASTag = Always on, Insurance = Weekly digest, Promotional = Monthly only.' → Reduces opt-out rate.

3. PREDICTIVE CROSS-SELL ENGINE

Priority: MEDIUM | Effort: HIGH | Impact: HIGH

CURRENT: Generic cross-sell based on transaction type. All users see same offers regardless of history.
RECOMMENDATION: Build ML model to predict optimal cross-sell timing and offer. Use vehicle age, service history, location, past behavior.
EXAMPLE: User's car is 3 years old, last service 6 months ago → App sends 'Time for annual service? 15% off at partner garage.' → Personalized.

Metrics

NORTH STAR: **LTV to CAC Ratio / ARPU** (Target: 3:1 ratio)

$$CAC = \frac{\text{Total Sales and Marketing Expenses}}{\text{Total Number of New Customers Acquired}}$$

$$LTV = ARPU \times \text{Gross Margin \%} \times \text{Customer Lifespan} \left(\text{or } \frac{1}{\text{Churn Rate}}\right)$$

$$ARPU = \frac{\text{Total Revenue}}{\text{Total Number of Users}}$$

L1 METRICS (Core Funnel)

Utility-to-Commerce Conversion

% of users buying services post-utility task

Measures thesis: can utility users convert?

Target: Double current rate

Service Retention Rate

Month-over-month renewal of car wash, insurance

Sustained value beyond first transaction

Target: 70% MoM retention

Time-to-Value

Time from app open to utility task completion

Speed is the core differentiator

Target: <30 seconds

L2 METRICS (Engagement)

Engagement Time Split

Time spent in marketplace UI vs utility UI

Higher discovery time = successful pivot

Target: 40% in discovery

Session Frequency

App opens without external SMS triggers

Organic opens = habit formation

Target: 3x/week organic

Cross-Sell Acceptance

% of users who engage with post-utility offers

Context-aware recommendation success

Target: 25% acceptance

COUNTER / FAILURE METRIC: Opt-out Rate — % of users disabling notifications. High opt-out indicates cross-sell fatigue. If opt-out exceeds 15%, pause all promotional notifications and redesign approach.

THANK YOU!